DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

6 **(a)** A car windscreen is covered in condensation (small droplets of water). Thermal energy is used to remove the droplets of water. The thermal energy is provided by three resistors on the windscreen.

Fig. 6.1 shows two possible circuits for the three resistors.

The three resistors are identical.

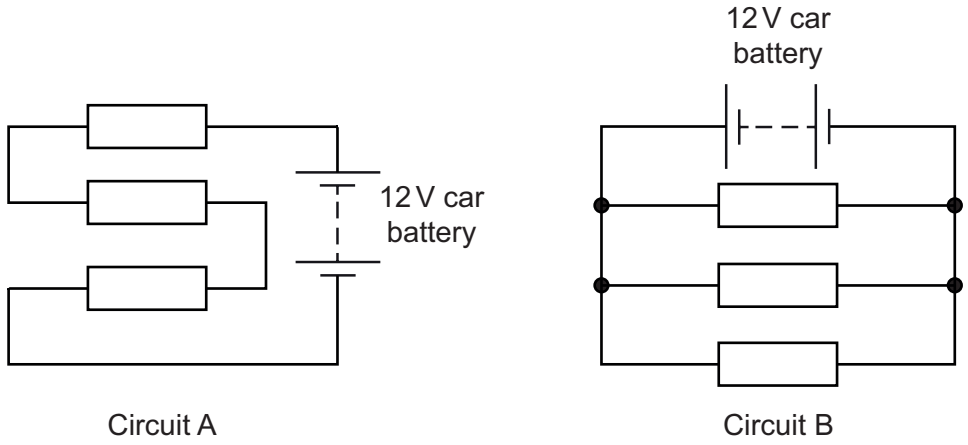

Fig. 6.1

(i) Describe **two** advantages of using Circuit B.

1 .....

.....

2 .....

.....

[2]

(ii) Describe, in terms of the water particles, the process by which the water droplets are removed from the car windscreen using the heater.

.....

.....

.....

..... [2]

(b) Fig. 6.2 shows a circuit containing two resistors, P and Q. The circuit is powered by a 12 V battery.

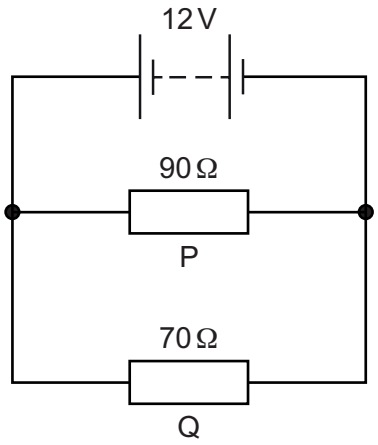

Fig. 6.2

(i) Calculate the current in resistor Q.

current = ... [2]

(ii) Calculate the energy transferred electrically when the current calculated in **(b)(i)** is present in resistor Q for 5 minutes.

energy = ... [3]

(iii) Energy is transferred from the battery by the electrical current.

State the energy store in the battery.

..... [1]

(iv) Calculate the total resistance of the circuit.

total resistance = ... [2]

[Total: 12]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN